Qualification Pack

Entrepreneurship in Solar Water Heater Manufacturing

QP Code: IID/ESS/Q0701

Version: 1.0

NSQF Level: 3.5

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Contents

IID/ESS/Q0701: Entrepreneurship in Solar Water Heater Manufacturing

Brief Job Description

The qualification for Entrepreneurship in Solar Water Heater Manufacturing is designed to equip individuals with the essential skills and knowledge to establish and manage their own small enterprise in the solar water heater manufacturing sector. It focuses on the technical aspects of solar water heater production, including system components, assembly, and quality standards, along with key entrepreneurial competencies such as business planning, marketing, financial management, and supply chain operations. The program also introduces learners to regulatory frameworks, intellectual property protection, and government initiatives promoting renewable energy ventures, enabling them to build sustainable and efficient solar water heater businesses.

Personal Attributes

Candidate should have Knowledge of solar energy systems and heating technologies, Visionary Thinking, Resilience and Perseverance, Problem-Solving and Analytical Skills, Leadership and Team Collaboration, Networking Skills, Communication and Interpersonal Skills, etc.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. IID/ESS/N0701: Entrepreneurship in Solar Water Heater Manufacturing

Qualification Pack (QP) Parameters

Sector	Environmental Science
Sub-Sector	
Occupation	Renewable Energy
Country	India
NSQF Level	3.5
Credits	3
Aligned to NCO/ISCO/ISIC Code	NCO 2015/1120.1700

Minimum Educational Qualification & Experience	11th Class (Pass) with 1 Year of experience OR 10th Class (Pass) with 1.5 years of experience In relevant Industry OR 8th Class pass with 4.5 years of experience In relevant Industry OR Previous relevant Qualification of NSQF Level (3 in relevant field) with 1.5 years of experience In relevant Industry
Minimum Level of Education for Training in School	8th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	17/12/2027
NSQC Approval Date	17/12/2024
Version	1.0
Reference code on NQR	NG-3.5-ES-03447-2024-V1-SS
NQR Version	1.0

IID/ESS/N0701: Entrepreneurship in Solar Water Heater Manufacturing

Description

The qualification for Entrepreneurship in Solar Water Heater Manufacturing is designed to equip individuals with the essential skills and knowledge to establish and manage their own small enterprise in the solar water heater manufacturing sector. It focuses on the technical aspects of solar water heater production, including system components, assembly, and quality standards, along with key entrepreneurial competencies such as business planning, marketing, financial management, and supply chain operations. The program also introduces learners to regulatory frameworks, intellectual property protection, and government initiatives promoting renewable energy ventures, enabling them to build sustainable and efficient solar water heater businesses.

Scope

The scope covers the following :

- Gain foundational knowledge in solar energy, radiation, solar-thermal principles, and their application in solar water heater technology.
- Understand machine, equipment, and infrastructure requirements for solar water heater production and optimization techniques.
- Acquire basic machine maintenance and calibration techniques to ensure consistent product quality. Understand the importance of insulation materials, temperature control, and regulation for efficient solar water heater performance.
- Create procurement strategies for sourcing raw materials and components, and ensure a reliable supply chain for manufacturing needs.
- Identify market trends and opportunities in the solar water heater manufacturing business and develop effective sales and marketing strategies.
- Develop strategies to forecast sales and plan for demand in the solar water heater manufacturing business to optimize inventory and resource allocation.

Elements and Performance Criteria

Introduction to Solar Energy, Radiation, Solar-Thermal Principles, and its Application

To be competent, the user/individual on the job must be able to:

PC1. Understand the basic concepts and significance of solar energy.

PC2. Factors affecting solar radiation: atmospheric conditions, geographic location, time of day/year.

PC3. Basic concepts of heat transfer: conduction, convection, and radiation.

PC4. Applications of Solar-Thermal Energy

Introduction to Types of Solar Collectors

To be competent, the user/individual on the job must be able to:

PC5. Understand the fundamental principles of solar collectors

PC6. Study about the different types of solar collectors and their operating mechanisms.

PC7. Explore the efficiency and performance parameters of each collector type.

PC8. Identify the applications and suitability of various solar collectors.

Working Principle of Solar Water Heater and Its Classification

To be competent, the user/individual on the job must be able to:

PC9. Understand the basic working principles of solar water heaters

PC10. Learn about the different classifications of solar water heaters

PC11. Explore the design, operation, and efficiency of various solar water heater types.

PC12. Identify the appropriate applications for each type of solar water heater

Understanding the Importance of Heat transfer Fluids and their types

To be competent, the user/individual on the job must be able to:

PC13. Understand the fundamental principles of heat transfer and the critical role of heat transfer fluids.

PC14. Learn about the different types of heat transfer fluids and their properties

PC15. Explore the criteria for selecting appropriate heat transfer fluids for specific applications.

PC16. Analyze the performance and efficiency implications of various heat transfer fluids

Introduction to Solar Water Heater Assembly Components and Possible Design Types

To be competent, the user/individual on the job must be able to:

PC17. Identify the key components: solar collectors, storage tanks, heat exchangers, controllers, and circulation pumps, etc

PC18. Understand the integration and interaction of components within the system.

PC19. Explore the design types of solar water heater.

PC20. Identify the steps in assembling a solar water heater.

Selection Criteria for Solar Water Heater

To be competent, the user/individual on the job must be able to:

PC21. Learn the techniques to evaluate different solar water heating systems based on performance metrics.

PC22. Explore the impact of environmental and site-specific conditions on system choice

PC23. Identify the system design and configuration based on specific application needs

PC24. Identify the system design and configuration based on specific application needs

Solar Water Heater Raw Materials

To be competent, the user/individual on the job must be able to:

PC25. Understand the key raw materials used in solar water heater components.

PC26. Learn about the properties and selection criteria for these materials.

PC27. Explore the impact of material choices on system performance and longevity.

PC28. Identify the sustainability and environmental considerations in material selection.

Quality Standards and Inspection Associated to Raw Materials

To be competent, the user/individual on the job must be able to:

PC29. Understand the importance of quality standards in the selection of raw materials for solar water heaters

PC30. Learn about the specific standards and certifications applicable to solar water heater materials.

PC31. Explore the inspection and testing methods used to ensure material quality.

PC32. Documentation and traceability requirements for quality assurance

Machine, Equipment and Infrastructure Requirement for Solar Water Heater Manufacturing

To be competent, the user/individual on the job must be able to:

PC33. Understand the key machinery and equipment used in the manufacturing of solar water heaters

PC34. Learn about the infrastructure requirements for a solar water heater manufacturing facility.

PC35. Identify the methods of Site selection and facility planning.

PC36. Identify the best practices for setting up and maintaining a manufacturing facility.

Solar Water Heater Manufacturing Process

To be competent, the user/individual on the job must be able to:

PC37. Understand the complete manufacturing process of solar water heaters

PC38. Learn the specific steps involved in producing each component of a solar water heater.

PC39. Explore the technologies and equipment used in the manufacturing process

PC40. Analysis of successful manufacturing facilities

Basic Machine Maintenance and Calibration Techniques

To be competent, the user/individual on the job must be able to:

PC41. Understand the importance of regular machine maintenance and calibration.

PC42. Learn basic maintenance procedures and troubleshooting techniques.

PC43. Gain knowledge of calibration methods for various types of manufacturing equipment

PC44. Develop best practices for maintaining and calibrating equipment in a solar water heater manufacturing facility.

Finished Solar Water Heater Testing, Performance Reporting and Packaging Methods

To be competent, the user/individual on the job must be able to:

PC45. Understand the importance of testing and performance reporting for finished solar water heaters.

PC46. Learn about various testing methods to ensure product quality and reliability.

PC47. Identify and gain knowledge in compiling and interpreting performance reports.

PC48. Explore effective packaging methods to protect solar water heaters during transportation and storage.

Safety and Health Standards in Solar Water Heater Manufacturing

To be competent, the user/individual on the job must be able to:

PC49. Understand the importance of safety and health standards in manufacturing

PC50. Learn about key regulations and guidelines for workplace safety and health

PC51. Develop skills in risk assessment and the implementation of safety protocols.

PC52. Explore best practices for maintaining a safe and healthy work environment.

Troubleshoots Analysis in Solar Water Heater Manufacturing

To be competent, the user/individual on the job must be able to:

PC53. Understand the impact of unresolved issues on product quality and manufacturing efficiency.

PC54. Identify the common issues and their causes in solar water heater manufacturing process

PC55. Understand the techniques for identifying the underlying causes of problems

PC56. Explore effective solutions and preventive measures to address and avoid issues.

Solar Water Heater Mounting and Installation Techniques

To be competent, the user/individual on the job must be able to:

PC57. Understand the importance of proper mounting and installation of solar water heaters

PC58. Learn the techniques for assessing installation sites and selecting appropriate mounting systems.

PC59. Explore the methods in installing solar water heater components securely and efficiently.

PC60. Explore safety measures and best practices to ensure reliable operation.

Implementing Waste Management Techniques for Solar Water Heater Business

To be competent, the user/individual on the job must be able to:

PC61. Identify waste streams generated during solar water heater manufacturing and installation.

PC62. Manage and recycle materials like glass, metals, and plastics.

PC63. Develop methods for safe disposal of electronic components and batteries.

PC64. Collaborate with recycling facilities for handling production waste.

Identifying Market Trends and Opportunities in the Solar Water Heater Business

To be competent, the user/individual on the job must be able to:

PC65. Learn how to conduct market research to gather insights into customer preferences, industry trends, and competitive dynamics.

PC66. Analyzing demand drivers such as government policies, energy costs, and consumer preferences.

PC67. Analyze market research findings to identify opportunities for business growth, product innovation, and marketing strategy development

PC68. Use market research data to identify emerging market segments and opportunity.

Developing Effective Sales and Marketing Strategies for the Solar Water Heater Manufacturing Business

To be competent, the user/individual on the job must be able to:

PC69. Understand the unique challenges and opportunities in sales and marketing for the solar water heater manufacturing business.

PC70. Identifying target markets and customer segments based on demographic, geographic, and psychographic factors

PC71. Identify and develop a sales strategy tailored to the unique selling points and benefits of solar water heater products

PC72. Implement targeted marketing campaigns and messaging to reach and engage specific customer segments effectively.

Sales Forecasting and Demand Planning in the Solar Water Heater Manufacturing Business

To be competent, the user/individual on the job must be able to:

PC73. Understand the importance of sales forecasting and demand planning for effective inventory management and resource allocation.

PC74. Learn different techniques and models for sales forecasting and demand planning.

PC75. Develop skills in analyzing market trends, customer behaviour, and external factors affecting demand.

PC76. • Explore strategies for optimizing inventory levels, production schedules, and supply chain efficiency based on sales forecasts and demand
• projections

Developing Risk Management Strategies for the Solar Water Heater Manufacturing Business

To be competent, the user/individual on the job must be able to:

PC77. Understand the importance of risk management in the context of solar water heater manufacturing.

PC78. Learn different techniques and frameworks for identifying, assessing, and prioritizing risks.

PC79. Learn to develop risk mitigation strategies and contingency plans.

PC80. Explore strategies for integrating risk management into overall business planning and decision-making processes.

Developing Procurement Strategies for Raw Materials and Components in the Solar Water Heater Manufacturing Business

To be competent, the user/individual on the job must be able to:

PC81. Understand the role and importance of procurement in solar water heater manufacturing.

PC82. Learn different sourcing strategies and supplier selection criteria.

PC83. Develop skills in negotiating contracts, managing supplier relationships, and ensuring supply chain resilience.

PC84. Explore strategies for sustainable procurement and responsible sourcing practices.

Developing Strategies for Solar Water Heater Logistics and Inventory Management

To be competent, the user/individual on the job must be able to:

PC85. • Understand the importance of logistics and inventory management for ensuring timely delivery, minimizing costs, and optimizing supply chain
• efficiency

PC86. Learn different techniques and strategies for optimizing transportation, warehousing, and inventory control.

PC87. Develop skills in demand forecasting, inventory planning, and supply chain optimization

PC88. Explore strategies for improving supply chain visibility, responsiveness, and resilience.

Building and Maintaining Effective Distributor Relationships in the Solar Water Heater Manufacturing Business

To be competent, the user/individual on the job must be able to:

PC89. Understand the significance of distributor relationships in solar water heater manufacturing

PC90. Learn strategies for identifying, selecting, and onboarding distributors.

PC91. Develop skills in managing and motivating distributors to maximize sales performance.

PC92. Explore techniques for fostering long-term, mutually beneficial partnerships with distributors

Budgeting and Financial Planning Techniques in the Solar Water Heater Manufacturing Business

To be competent, the user/individual on the job must be able to:

PC93. Understand the importance of budgeting and financial planning in solar water heater manufacturing.

PC94. Learn different budgeting techniques and financial forecasting methods

PC95. Develop skills in analyzing financial statements, evaluating investment opportunities, and managing cash flow

PC96. Explore strategies for aligning financial planning with business objectives and maximizing profitability.

Revenue Management Techniques for the Solar Water Heater Manufacturing Business

To be competent, the user/individual on the job must be able to:

PC97. Understand the principles and importance of revenue management in solar water heater manufacturing.

PC98. Learn different revenue management techniques and pricing strategies.

PC99. Develop skills in market segmentation, demand forecasting, and pricing optimization

PC100. Explore strategies for aligning revenue management with business objectives and maximizing profitability.

Implementing Time Management Strategies in the Solar Water Heater Manufacturing Business

To be competent, the user/individual on the job must be able to:

PC101. Understand the importance of time management in solar water heater manufacturing

PC102. Learn different time management techniques and productivity strategies.

PC103. Develop skills in analyzing production processes, identifying inefficiencies, and implementing improvements.

PC104. Explore strategies for optimizing workflow, minimizing downtime, and meeting production schedules.

Human Resources Management in the Solar Water Heater Manufacturing Business

To be competent, the user/individual on the job must be able to:

PC105. Understand the importance of human resources management in solar water heater manufacturing Industry

PC106. Learn different HRM functions, including recruitment, training, performance management, and employee relations.

PC107. Develop skills in talent acquisition, workforce development, and employee engagement.

PC108. Explore strategies for building a high-performing and motivated workforce in the solar water heater manufacturing sector

Implementing Strategies for Export and International Market Development for the Solar Water Heater Manufacturing Business

To be competent, the user/individual on the job must be able to:

PC109. Understand the importance of export and international market development in solar water heater manufacturing.

PC110. Learn different strategies for market research, market entry, and market expansion.

PC111. Develop skills in identifying target markets, building distribution networks, and managing international trade relationships.

PC112. Explore strategies for overcoming challenges and capitalizing on opportunities in global markets.

Manage Branding and Promotion of Business

To be competent, the user/individual on the job must be able to:

PC113. Analyze the target audience to determine their preferences and behavior

PC114. Define the brand’s mission, vision, and unique selling proposition (USP).

PC115. Design a comprehensive branding strategy aligned with business objectives.

PC116. Utilize market research tools to assess competitors' branding and promotional strategies.

Detailed Project Report on Solar Water Heater Manufacturing Business

To be competent, the user/individual on the job must be able to:

PC117. Capability to plan and propose a project effectively.

PC118. Proficiency in selecting and utilizing appropriate research methods and collecting data.

PC119. Competence in analyzing and interpreting data accurately.
PC120. Ability to write and structure a comprehensive project report.
PC121. Understanding of the solar water heater market
PC122. Ability to manage financial projections and cash flow management.
PC123. Ability to interpret research findings and draw conclusions.
PC124. Presentation and communication skills to present project objectives, findings, and recommendations clearly.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Research and analysis
KU2. Economic and business terminology
KU3. Creative thinking techniques
KU4. Brainstorming and ideation
KU5. Innovation management
KU6. Project management
KU7. Business planning
KU8. Financial modeling
KU9. Market research
KU10. Risk assessment
KU11. Legal research
KU12. Documentation and filing
KU13. Local, state, and central regulations
KU14. Budgeting and financial management
KU15. Resourcefulness and ingenuity
KU16. Negotiation
KU17. Time management
KU18. Government programs and policies
KU19. Grant writing
KU20. Market analysis
KU21. Strategic planning
KU22. Benchmarking
KU23. Competitive analysis
KU24. Basic business management
KU25. Industry knowledge
KU26. Technical knowledge of car cleaning
KU27. Equipment handling
KU28. Safety protocols
KU29. Quality control

KU30. Surface preparation techniques
KU31. Use of cleaning products and equipment
KU32. Environmental regulations
KU33. Market segmentation
KU34. Customer profiling
KU35. Data analysis
KU36. Business model development
KU37. Business Compliance
KU38. Documentation and record-keeping
KU39. Digital marketing
KU40. Brand management
KU41. Social media marketing
KU42. Content creation
KU43. Business plan writing
KU44. Financial projections

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Critical Thinking
GS2. Communication Skills
GS3. Adaptability
GS4. Problem-solving
GS5. Networking
GS6. Creative Thinking
GS7. Open-mindedness
GS8. Collaboration
GS9. Curiosity
GS10. Risk-taking
GS11. Decision-making
GS12. Time Management
GS13. Perseverance
GS14. Resourcefulness
GS15. Resilience
GS16. Positive Attitude
GS17. Self-reflection
GS18. Stress Management
GS19. Flexibility
GS20. Attention to Detail
GS21. Research Skills

GS22. Patience
GS23. Negotiation Skills
GS24. Persuasion
GS25. Leadership
GS26. Analytical Thinking
GS27. Physical Stamina
GS28. Empathy
GS29. Social Media Skills
GS30. Creativity

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Solar Energy, Radiation, Solar-Thermal Principles, and its Application	**4**	**-**	**-**	**-**
PC1. Understand the basic concepts and significance of solar energy.	1	-	-	-
PC2. Factors affecting solar radiation: atmospheric conditions, geographic location, time of day/year.	1	-	-	-
PC3. Basic concepts of heat transfer: conduction, convection, and radiation.	1	-	-	-
PC4. Applications of Solar-Thermal Energy	1	-	-	-
Introduction to Types of Solar Collectors	**4**	**-**	**-**	**-**
PC5. Understand the fundamental principles of solar collectors	1	-	-	-
PC6. Study about the different types of solar collectors and their operating mechanisms.	1	-	-	-
PC7. Explore the efficiency and performance parameters of each collector type.	1	-	-	-
PC8. Identify the applications and suitability of various solar collectors.	1	-	-	-
Working Principle of Solar Water Heater and Its Classification	**4**	**-**	**-**	**-**
PC9. Understand the basic working principles of solar water heaters	1	-	-	-
PC10. Learn about the different classifications of solar water heaters	1	-	-	-
PC11. Explore the design, operation, and efficiency of various solar water heater types.	1	-	-	-
PC12. Identify the appropriate applications for each type of solar water heater	1	-	-	-
Understanding the Importance of Heat transfer Fluids and their types	**4**	**-**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Understand the fundamental principles of heat transfer and the critical role of heat transfer fluids.	1	-	-	-
PC14. Learn about the different types of heat transfer fluids and their properties	1	-	-	-
PC15. Explore the criteria for selecting appropriate heat transfer fluids for specific applications.	1	-	-	-
PC16. Analyze the performance and efficiency implications of various heat transfer fluids	1	-	-	-
Introduction to Solar Water Heater Assembly Components and Possible Design Types	**4**	**-**	**-**	**-**
PC17. Identify the key components: solar collectors, storage tanks, heat exchangers, controllers, and circulation pumps, etc	1	-	-	-
PC18. Understand the integration and interaction of components within the system.	1	-	-	-
PC19. Explore the design types of solar water heater.	1	-	-	-
PC20. Identify the steps in assembling a solar water heater.	1	-	-	-
Selection Criteria for Solar Water Heater	**4**	**-**	**-**	**-**
PC21. Learn the techniques to evaluate different solar water heating systems based on performance metrics.	1	-	-	-
PC22. Explore the impact of environmental and site-specific conditions on system choice	1	-	-	-
PC23. Identify the system design and configuration based on specific application needs	1	-	-	-
PC24. Identify the system design and configuration based on specific application needs	1	-	-	-
Solar Water Heater Raw Materials	**4**	**-**	**-**	**-**
PC25. Understand the key raw materials used in solar water heater components.	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. Learn about the properties and selection criteria for these materials.	1	-	-	-
PC27. Explore the impact of material choices on system performance and longevity.	1	-	-	-
PC28. Identify the sustainability and environmental considerations in material selection.	1	-	-	-
Quality Standards and Inspection Associated to Raw Materials	**4**	**-**	**-**	**-**
PC29. Understand the importance of quality standards in the selection of raw materials for solar water heaters	1	-	-	-
PC30. Learn about the specific standards and certifications applicable to solar water heater materials.	1	-	-	-
PC31. Explore the inspection and testing methods used to ensure material quality.	1	-	-	-
PC32. Documentation and traceability requirements for quality assurance	1	-	-	-
Machine, Equipment and Infrastructure Requirement for Solar Water Heater Manufacturing	**4**	**-**	**-**	**-**
PC33. Understand the key machinery and equipment used in the manufacturing of solar water heaters	1	-	-	-
PC34. Learn about the infrastructure requirements for a solar water heater manufacturing facility.	1	-	-	-
PC35. Identify the methods of Site selection and facility planning.	1	-	-	-
PC36. Identify the best practices for setting up and maintaining a manufacturing facility.	1	-	-	-
Solar Water Heater Manufacturing Process	**4**	**-**	**-**	**-**
PC37. Understand the complete manufacturing process of solar water heaters	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC38. Learn the specific steps involved in producing each component of a solar water heater.	1	-	-	-
PC39. Explore the technologies and equipment used in the manufacturing process	1	-	-	-
PC40. Analysis of successful manufacturing facilities	1	-	-	-
Basic Machine Maintenance and Calibration Techniques	**4**	**-**	**-**	**-**
PC41. Understand the importance of regular machine maintenance and calibration.	1	-	-	-
PC42. Learn basic maintenance procedures and troubleshooting techniques.	1	-	-	-
PC43. Gain knowledge of calibration methods for various types of manufacturing equipment	1	-	-	-
PC44. Develop best practices for maintaining and calibrating equipment in a solar water heater manufacturing facility.	1	-	-	-
Finished Solar Water Heater Testing, Performance Reporting and Packaging Methods	**4**	**-**	**-**	**-**
PC45. Understand the importance of testing and performance reporting for finished solar water heaters.	1	-	-	-
PC46. Learn about various testing methods to ensure product quality and reliability.	1	-	-	-
PC47. Identify and gain knowledge in compiling and interpreting performance reports.	1	-	-	-
PC48. Explore effective packaging methods to protect solar water heaters during transportation and storage.	1	-	-	-
Safety and Health Standards in Solar Water Heater Manufacturing	**4**	**-**	**-**	**-**
PC49. Understand the importance of safety and health standards in manufacturing	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC50. Learn about key regulations and guidelines for workplace safety and health	1	-	-	-
PC51. Develop skills in risk assessment and the implementation of safety protocols.	1	-	-	-
PC52. Explore best practices for maintaining a safe and healthy work environment.	1	-	-	-
Troubleshoots Analysis in Solar Water Heater Manufacturing	**4**	**-**	**-**	**-**
PC53. Understand the impact of unresolved issues on product quality and manufacturing efficiency.	1	-	-	-
PC54. Identify the common issues and their causes in solar water heater manufacturing process	1	-	-	-
PC55. Understand the techniques for identifying the underlying causes of problems	1	-	-	-
PC56. Explore effective solutions and preventive measures to address and avoid issues.	1	-	-	-
Solar Water Heater Mounting and Installation Techniques	**4**	**-**	**-**	**-**
PC57. Understand the importance of proper mounting and installation of solar water heaters	1	-	-	-
PC58. Learn the techniques for assessing installation sites and selecting appropriate mounting systems.	1	-	-	-
PC59. Explore the methods in installing solar water heater components securely and efficiently.	1	-	-	-
PC60. Explore safety measures and best practices to ensure reliable operation.	1	-	-	-
Implementing Waste Management Techniques for Solar Water Heater Business	**4**	**-**	**-**	**-**
PC61. Identify waste streams generated during solar water heater manufacturing and installation.	1	-	-	-
PC62. Manage and recycle materials like glass, metals, and plastics.	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC63. Develop methods for safe disposal of electronic components and batteries.	1	-	-	-
PC64. Collaborate with recycling facilities for handling production waste.	1	-	-	-
Identifying Market Trends and Opportunities in the Solar Water Heater Business	**4**	**-**	**-**	**-**
PC65. Learn how to conduct market research to gather insights into customer preferences, industry trends, and competitive dynamics.	1	-	-	-
PC66. Analyzing demand drivers such as government policies, energy costs, and consumer preferences.	1	-	-	-
PC67. Analyze market research findings to identify opportunities for business growth, product innovation, and marketing strategy development	1	-	-	-
PC68. Use market research data to identify emerging market segments and opportunity.	1	-	-	-
Developing Effective Sales and Marketing Strategies for the Solar Water Heater Manufacturing Business	**4**	**-**	**-**	**-**
PC69. Understand the unique challenges and opportunities in sales and marketing for the solar water heater manufacturing business.	1	-	-	-
PC70. Identifying target markets and customer segments based on demographic, geographic, and psychographic factors	1	-	-	-
PC71. Identify and develop a sales strategy tailored to the unique selling points and benefits of solar water heater products	1	-	-	-
PC72. Implement targeted marketing campaigns and messaging to reach and engage specific customer segments effectively.	1	-	-	-
Sales Forecasting and Demand Planning in the Solar Water Heater Manufacturing Business	**4**	**-**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC73. Understand the importance of sales forecasting and demand planning for effective inventory management and resource allocation.	1	-	-	-
PC74. Learn different techniques and models for sales forecasting and demand planning.	1	-	-	-
PC75. Develop skills in analyzing market trends, customer behaviour, and external factors affecting demand.	1	-	-	-
PC76. • Explore strategies for optimizing inventory levels, production schedules, and supply chain efficiency based on sales forecasts and demand • projections	1	-	-	-
Developing Risk Management Strategies for the Solar Water Heater Manufacturing Business	**4**	**-**	**-**	**-**
PC77. Understand the importance of risk management in the context of solar water heater manufacturing.	1	-	-	-
PC78. Learn different techniques and frameworks for identifying, assessing, and prioritizing risks.	1	-	-	-
PC79. Learn to develop risk mitigation strategies and contingency plans.	1	-	-	-
PC80. Explore strategies for integrating risk management into overall business planning and decision-making processes.	1	-	-	-
Developing Procurement Strategies for Raw Materials and Components in the Solar Water Heater Manufacturing Business	**4**	**-**	**-**	**-**
PC81. Understand the role and importance of procurement in solar water heater manufacturing.	1	-	-	-
PC82. Learn different sourcing strategies and supplier selection criteria.	1	-	-	-
PC83. Develop skills in negotiating contracts, managing supplier relationships, and ensuring supply chain resilience.	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC84. Explore strategies for sustainable procurement and responsible sourcing practices.	1	-	-	-
Developing Strategies for Solar Water Heater Logistics and Inventory Management	**4**	**-**	**-**	**-**
PC85. • Understand the importance of logistics and inventory management for ensuring timely delivery, minimizing costs, and optimizing supply chain • efficiency	1	-	-	-
PC86. Learn different techniques and strategies for optimizing transportation, warehousing, and inventory control.	1	-	-	-
PC87. Develop skills in demand forecasting, inventory planning, and supply chain optimization	1	-	-	-
PC88. Explore strategies for improving supply chain visibility, responsiveness, and resilience.	1	-	-	-
Building and Maintaining Effective Distributor Relationships in the Solar Water Heater Manufacturing Business	**4**	**-**	**-**	**-**
PC89. Understand the significance of distributor relationships in solar water heater manufacturing	1	-	-	-
PC90. Learn strategies for identifying, selecting, and onboarding distributors.	1	-	-	-
PC91. Develop skills in managing and motivating distributors to maximize sales performance.	1	-	-	-
PC92. Explore techniques for fostering long-term, mutually beneficial partnerships with distributors	1	-	-	-
Budgeting and Financial Planning Techniques in the Solar Water Heater Manufacturing Business	**4**	**-**	**-**	**-**
PC93. Understand the importance of budgeting and financial planning in solar water heater manufacturing.	1	-	-	-
PC94. Learn different budgeting techniques and financial forecasting methods	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC95. Develop skills in analyzing financial statements, evaluating investment opportunities, and managing cash flow	1	-	-	-
PC96. Explore strategies for aligning financial planning with business objectives and maximizing profitability.	1	-	-	-
Revenue Management Techniques for the Solar Water Heater Manufacturing Business	**4**	**-**	**-**	**-**
PC97. Understand the principles and importance of revenue management in solar water heater manufacturing.	1	-	-	-
PC98. Learn different revenue management techniques and pricing strategies.	1	-	-	-
PC99. Develop skills in market segmentation, demand forecasting, and pricing optimization	1	-	-	-
PC100. Explore strategies for aligning revenue management with business objectives and maximizing profitability.	1	-	-	-
Implementing Time Management Strategies in the Solar Water Heater Manufacturing Business	**4**	**-**	**-**	**-**
PC101. Understand the importance of time management in solar water heater manufacturing	1	-	-	-
PC102. Learn different time management techniques and productivity strategies.	1	-	-	-
PC103. Develop skills in analyzing production processes, identifying inefficiencies, and implementing improvements.	1	-	-	-
PC104. Explore strategies for optimizing workflow, minimizing downtime, and meeting production schedules.	1	-	-	-
Human Resources Management in the Solar Water Heater Manufacturing Business	**4**	**-**	**-**	**-**
PC105. Understand the importance of human resources management in solar water heater manufacturing Industry	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC106. Learn different HRM functions, including recruitment, training, performance management, and employee relations.	1	-	-	-
PC107. Develop skills in talent acquisition, workforce development, and employee engagement.	1	-	-	-
PC108. Explore strategies for building a high-performing and motivated workforce in the solar water heater manufacturing sector	1	-	-	-
Implementing Strategies for Export and International Market Development for the Solar Water Heater Manufacturing Business	**4**	**-**	**-**	**-**
PC109. Understand the importance of export and international market development in solar water heater manufacturing.	1	-	-	-
PC110. Learn different strategies for market research, market entry, and market expansion.	1	-	-	-
PC111. Develop skills in identifying target markets, building distribution networks, and managing international trade relationships.	1	-	-	-
PC112. Explore strategies for overcoming challenges and capitalizing on opportunities in global markets.	1	-	-	-
Manage Branding and Promotion of Business	**4**	**-**	**-**	**-**
PC113. Analyze the target audience to determine their preferences and behavior	1	-	-	-
PC114. Define the brand's mission, vision, and unique selling proposition (USP).	1	-	-	-
PC115. Design a comprehensive branding strategy aligned with business objectives.	1	-	-	-
PC116. Utilize market research tools to assess competitors' branding and promotional strategies.	1	-	-	-
Detailed Project Report on Solar Water Heater Manufacturing Business	**-**	**-**	**48**	**16**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC117. Capability to plan and propose a project effectively.	-	-	6	2
PC118. Proficiency in selecting and utilizing appropriate research methods and collecting data.	-	-	6	2
PC119. Competence in analyzing and interpreting data accurately.	-	-	6	2
PC120. Ability to write and structure a comprehensive project report.	-	-	6	2
PC121. Understanding of the solar water heater market	-	-	6	2
PC122. Ability to manage financial projections and cash flow management.	-	-	6	2
PC123. Ability to interpret research findings and draw conclusions.	-	-	6	2
PC124. Presentation and communication skills to present project objectives, findings, and recommendations clearly.	-	-	6	2
NOS Total	**116**	**-**	**48**	**16**

National Occupational Standards (NOS) Parameters

NOS Code	IID/ESS/N0701
NOS Name	Entrepreneurship in Solar Water Heater Manufacturing
Sector	Environmental Science
Sub-Sector	
Occupation	Renewable Energy
NSQF Level	3.5
Credits	3
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment Overview: The purpose of this assessment SOP is to provide a structured and transparent process for evaluating students for the qualification "Entrepreneurship in Solar Water Heater

Manufacturing Business." This process aims to identify candidates who possess the skills, knowledge, and potential to excel in this field, ensuring that only the most suitable students are selected.

To achieve this, the assessment will be conducted through a multi-faceted approach, including a written assessment, case study analysis, MCQ based segment-wise final tests, development, and evaluation

of a comprehensive project report based on experiential learning, and a viva voice examination on the project work undertaken by the students.

Assessment Methods: The assessment for the online course "Entrepreneurship in Solar Water Heater Business" involves a multi-faceted approach to thoroughly evaluate candidates' qualifications and

suitability. The key methods include:

• Written Assessment: The written assessment is an online test designed to evaluate candidates' knowledge of solar water heater technology, business concepts, and industry trends. This test emphasizes technical understanding, analytical skills, and theoretical knowledge relevant to the course. Candidates are required to complete written assignments and case studies simulating scenarios

encountered in the solar water heater industry, which assess their critical thinking, problem-solving, and decision-making skills.

• Case Study Analysis: Candidates are provided with real-world scenarios related to solar water heater entrepreneurship. They are expected to analyze these cases, identify problems, propose

solutions, and demonstrate their problem-solving and decision-making skills. This method assesses the practical application of theoretical knowledge and entrepreneurial thinking.

• MCQ-Based Segment-Wise Final Tests: Multiple-choice questions (MCQs) are used to test candidates' knowledge and understanding across different segments of the course. These tests are

structured to evaluate comprehension of key concepts, retention of information, and the ability to apply knowledge in various contexts. The final test comprises MCQs covering each module of the

qualification, assessing candidates' overall knowledge and comprehension of the course material.

• Development and Evaluation of a Comprehensive Project Report: Candidates must develop a detailed project report based on experiential learning, involving real-world application, research,

planning, and execution related to solar water heater manufacturing and entrepreneurship. The project report is evaluated for originality, depth of analysis, feasibility of the business plan, and

practical insights. This assessment measures candidates’ ability to apply theoretical knowledge in a practical context.

• Viva Voce Examination on the Project Work: The viva voce is an oral examination that assesses candidates' understanding of their project

Minimum Aggregate Passing % at QP Level : 50

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
IID/ESS/N0701.Entrepreneurship in Solar Water Heater Manufacturing	116	-	48	16	180	100

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	116	0	48	16	180	100

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.